Electronic Devices and Circuits Lab

5-BIT SUCCESSIVE APPROXIMATION ADC



Avyaaz - EE23BTECH11013

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5-Bit Successive Approximation ADC

1.1 Objective

The objective of this experiment is to demonstrate the SAR ADC which converts an analog input signal into a corresponding 5-bit digital output by implementing a binary search algorithm.

1.2 Components

The main components in a 5-bit SAR ADC are as follows:

- Comparator ×1
- Digital-to-Analog Converter (DAC) ×1
- Successive Approximation Register (SAR) ×1
- Sample and Hold Circuit ×1

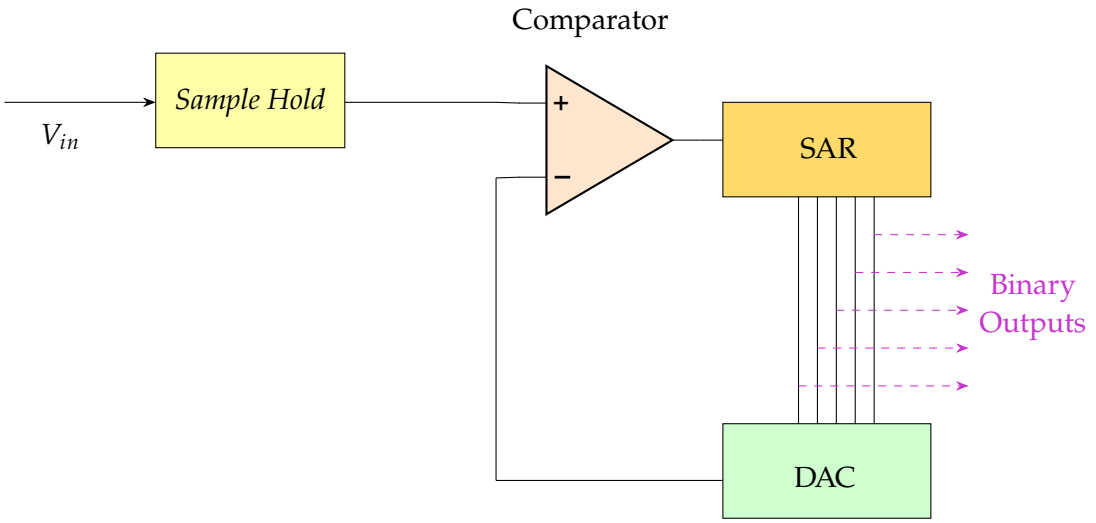


Figure 1.1: *Successive Approximation (SAR) ADC*

1.3 Working Principle

1. **Sample-Hold Circuit**
   A Sample and Hold circuit captures the input voltage at a specific moment and holds this value constant for a defined period. The time when the circuit captures the input voltage is called the sampling time, while the duration for which the value remains fixed is referred to as the holding time. When the input signal is applied, the first op-amp sends this signal to the drain of the MOSFET, which acts as a switch controlled by a control voltage. When the control voltage is high, the circuit closes, allowing the capacitor to start charging. When the control voltage is low, the circuit opens, and the capacitor holds the charge until the control voltage goes high again.

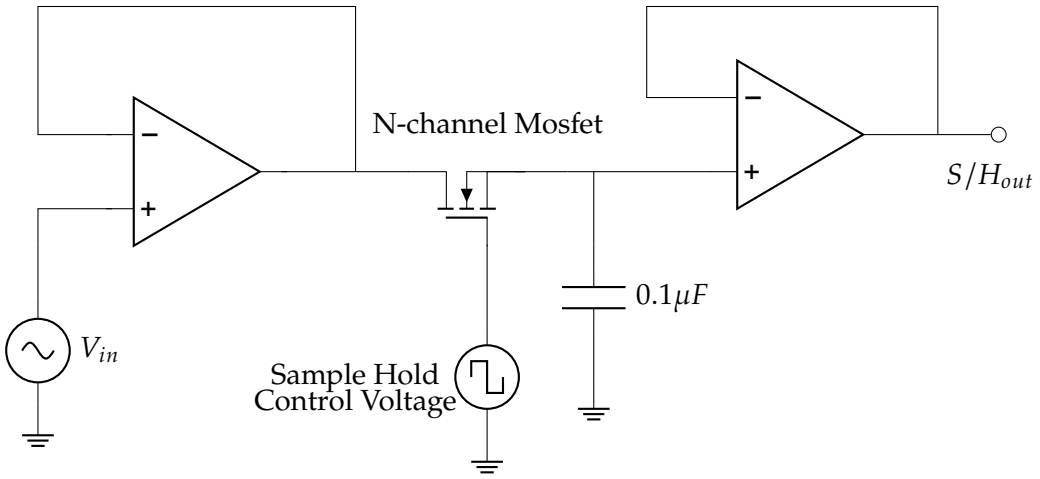
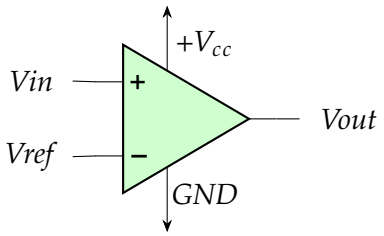


Figure 1.2: *Sample Hold Circuit*

2. **Comparator**
   The output of the sample and hold circuit is connected to the +ve terminal of the comparator, while the output of the DAC is connected to the -ve terminal. The comparator calculates the as shown below and sends to SAR.



$V_{in} > V_{Ref}$ then $V_{out} = +V_{cc}$
$V_{in} < V_{Ref}$ then $V_{out} = \text{GND}$

3. **Successive Approximation Register**
   - In the 5-bit SAR, initially, the D input of the upper-left flip-flop is connected to ground, with ($\overline{\text{CLR}}$) high and ($\overline{\text{PRE}}$) low. Since the enable pin is low, the

output Q is '1'. The $\overline{Q}$ output is then passed to the $\overline{\text{PRE}}$ of the flip-flop below it. With $\overline{\text{CLR}}$ low, this flip-flop outputs a garbage value until the enable pin becomes high.

- For the other upper flip-flops, $\overline{\text{PRE}}$ is high and $\overline{\text{CLR}}$ is low, it outputs $Q = 0$. Then, the $\overline{Q}$ output is passed to the $\overline{\text{PRE}}$ input of the corresponding flip-flop below. With $\overline{\text{CLR}}$ still low, this flip-flop outputs '0'. Resulting the SAR output initializing at '10000'.
- When the enable signal goes high, the MSB is set to '1', and the other flip-flop outputs retain their previous states because the clock (CLK) is 0.
- Considering an example with a DC analog input voltage of 21V, the initialized SAR output of '10000' (representing 16V) is less than the input signal. This results in the comparator output being logic high. In the next clock cycle, the upper-left flip-flop's output, which was '1', is passed to the next flip-flop, setting its output to '1' as well. The $\overline{Q} = 0$ output is passed to the $\overline{\text{PRE}}$ input of the flip-flop below, resulting in an output of '1' for this flip-flop. This value is then passed back to the previous flip-flop as CLK, ensuring it accepts the comparator output as input and sets D_4 to '1'.
- Now, the SAR output '11000' (representing 24V) is greater than the input voltage, making the comparator output logic low. In the next clock cycle, the output from the second upper flip-flop is passed to the third flip-flop, setting its output to '1' as well. The $\overline{Q} = 0$ output is passed to the $\overline{\text{PRE}}$ input of the flip-flop below, resulting in an output of '1' for this flip-flop. This value is then passed back to the previous flip-flop as the clock input, ensuring that it takes the comparator output as input and sets D_3 to '0'.

INPUTS				**OUTPUTS**	
$\overline{\textbf{PRE}}$	$\overline{\textbf{CLR}}$	**CLK**	**D**	**Q**	$\overline{\textbf{Q}}$
0	1	X	X	1	0
1	0	X	X	0	1
0	0	X	X	X	X
1	1	↑	1	1	0
1	1	↑	0	0	1
1	1	0	X	Q_0	$\overline{Q_0}$

Table 1.1: *Truth Table of the D-FlipFlops*

- As this process continues, each upper flip-flop passes a '1' in its respective clock cycle and subsequently passes $\overline{Q} = 0$ to the flip-flop below it. This allows the previous flip-flops to receive the clock signal, enabling them to accept the comparator output as input & setting their output. After the complete conversion process, the SAR output aligns with input analog signal.
- For a continuous analog signal, the enable signal needs to be held low briefly to reset the binary outputs to '10000' for each new analog input, as the binary search begins from '10000'. Then, the enable pin is set high to start the

conversion process.

	Sequencer Register Outputs						Code Register Outputs					Comparator Output
	DFF4	DFF3	DFF2	DFF1	DFF0	DFF_{end}	DFF_4	DFF_3	DFF_2	DFF_1	DFF_0	
0	1	0	0	0	0	0	0	0	0	0	0	X
1	0	1	0	0	0	0	1	0	0	0	0	B4
2	0	0	1	0	0	0	B4	1	0	0	0	B3
3	0	0	0	1	0	0	B4	B3	1	0	0	B2
4	0	0	0	0	1	0	B4	B3	B2	1	0	B1
5	0	0	0	0	0	1	B4	B3	B2	B1	1	B0
6	0	0	0	0	0	0	B4	B3	B2	B1	B0	X

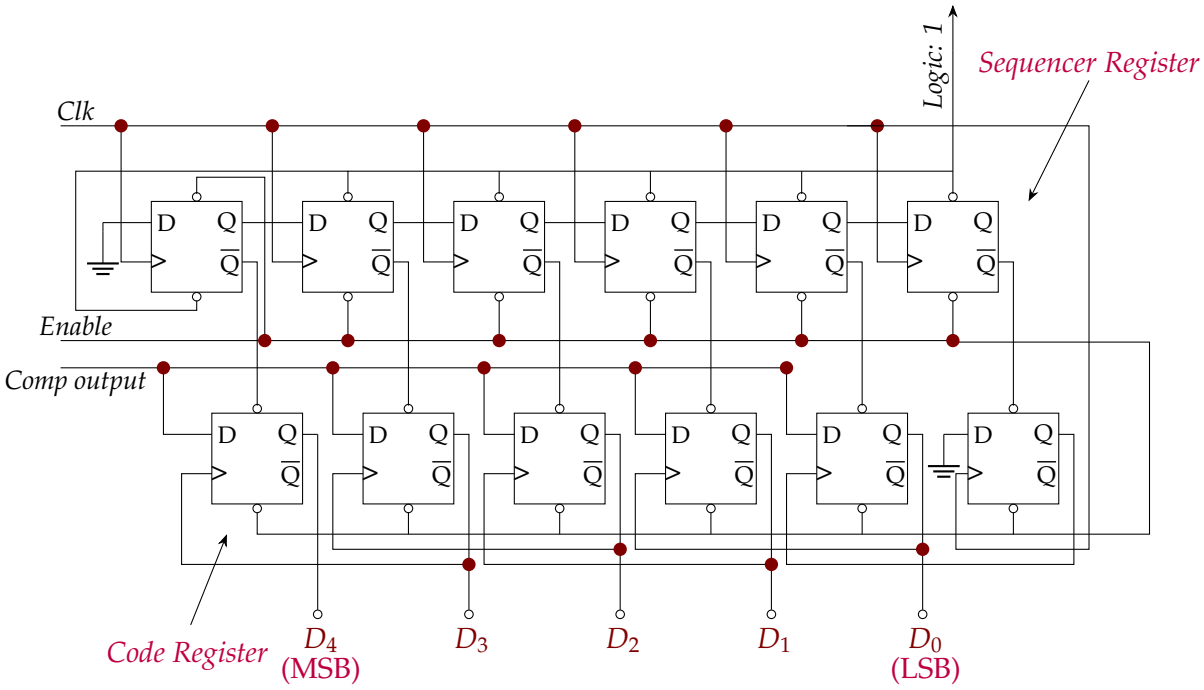


Figure 1.3: *Successive Approximation Register Control Logic*

4. **Digital to Analog Converter**
   A Digital-to-Analog Converter (DAC) converts digital output into corresponding analog voltages.

   - The R-2R ladder DAC consists of a network of resistors arranged in a ladder-like structure, using only two resistor values R and 2R to create a binary-weighted analog output. The R-2R ladder DAC operates on the principle of current division, where each bit in the binary input controls a switch connected to either a reference voltage V_{ref} (typically +V) or ground (0V).
   - Each bit in the input binary number controls whether its corresponding resistor branch connects to V_{ref} or ground:

– The most significant bit (MSB) contributes the largest portion to the output voltage.
– Each subsequent bit contributes half as much as the previous bit, following a binary weighting.

The analog output voltage V_{DAC} is calculated by summing the contributions of each bit:

$$V_{DAC} = \frac{V_{D_0} + 2V_{D_1} + 4V_{D_2} + 8V_{D_3} + 16V_{D_4}}{2^5}$$

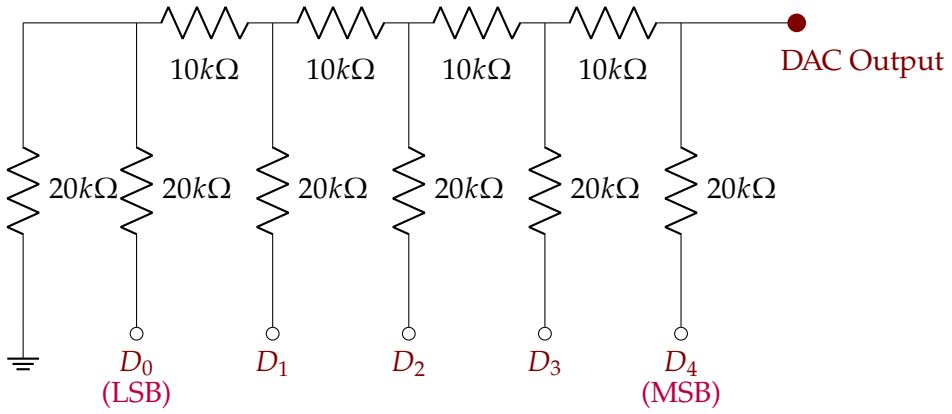


Figure 1.4: *DAC using R-2R Ladder*

1.4 Procedure

Assemble the SAR ADC circuit by connecting the components, including the comparator, Digital-to-Analog Converter (DAC), D flip-flops, sample and hold circuit, and successive approximation register (SAR) logic, as per the circuit diagram (Fig. 1.1). After ensuring connections, supply a stable analog input voltage to the sample and hold circuit.

Once the setup was complete, the SAR logic was triggered to start the binary search algorithm, beginning with the most significant bit (MSB) and iterating through each bit position to approximate the analog input value. After the conversion process completes, the final 5-bit digital output will be recorded as the binary representation of the input voltage.

This procedure is repeated for various analog input voltages to confirm the accuracy and functionality of the SAR ADC across a range of values. Finally, the recorded digital outputs are compared with expected values. For continuous analog input signals, you can use sinusoidal or ramp waveforms. The SAR ADC will then produce binary outputs ranging from '00000' to '11111'. When using a 5V reference, this range represents analog input values from 0V to 5V.

1.5 Results

The plot (Fig. 1.5) below shows the output of a SAR ADC. The data was collected by reading the SAR output using the digital pins of an Arduino. These digital values were then converted to decimal format and plotted using Python.

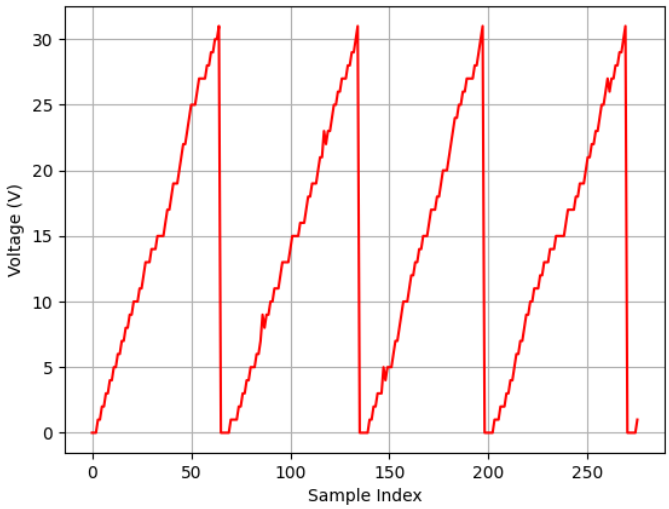


Figure 1.5: *Plot of SAR ADC*

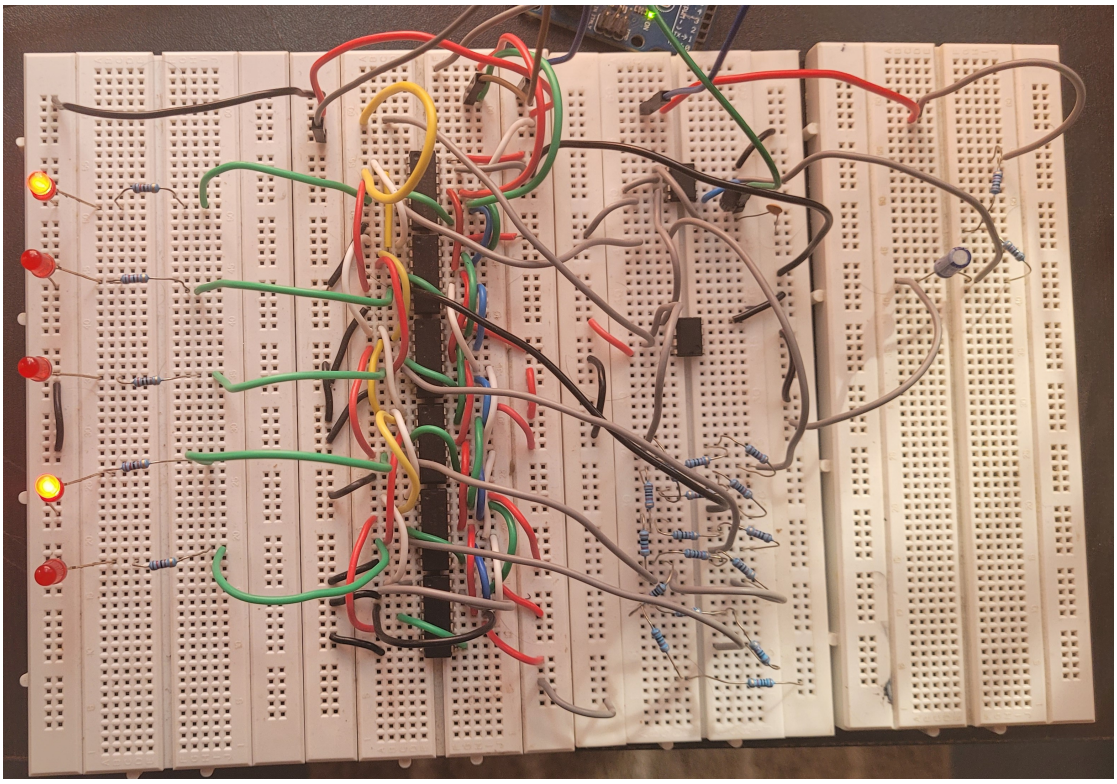


Figure 1.6: *Circuit Setup of a 5-bit Successive Approximation Register (SAR) ADC with an 18V.*

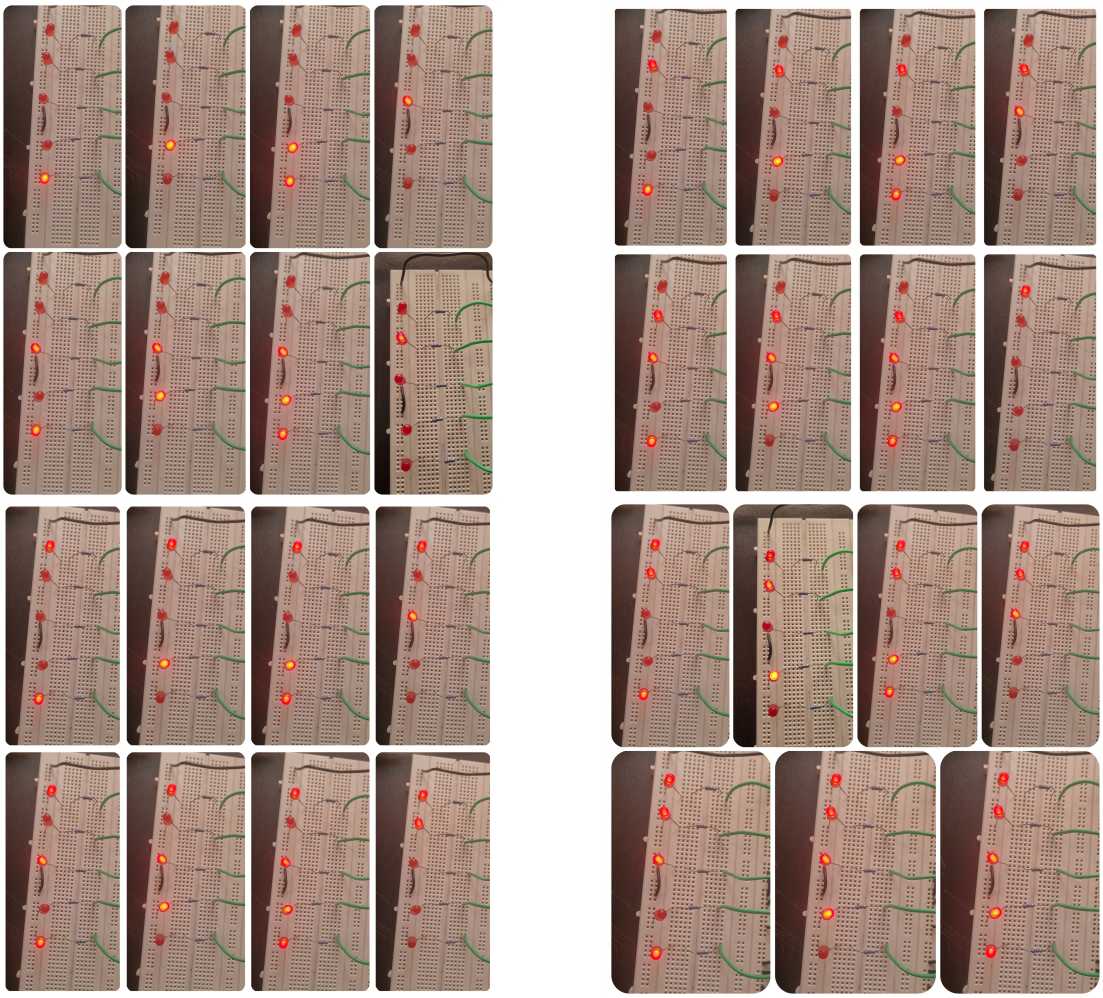


Figure 1.7: *Different outputs of the 5-bit SAR ADC operation.*

1.6 Conclusion

The 5-bit SAR ADC is an efficient method for analog-to-digital conversion using a binary search algorithm. It performs a step-by-step comparison of the analog input by determining each bit of the digital output. This experiment demonstrates the the sample-and-hold circuit, comparator, SAR logic, and the DAC.

1.7 Precautions

1. Use a stable clock to avoid timing errors.
2. Ensure SAR logic timing is accurate for correct sampling and comparisons.
3. Use a stable power supply within the range of LM358.
4. Ensure DAC can settle quickly enough in each cycle.
5. Use a sample-and-hold circuit with fast acquisition and hold times.